

Transport–entropy inequalities for Connes–Rieffel metrics

Solution packet for `fa_banach_001`

11 August 2026

Status. Scoped full resolution of the transport question in the finite tracial framework of Junge–Zeng: an exact necessary-and-sufficient exponential-moment criterion for an arbitrary seminorm L , an automatic positive answer for finite-diameter (hence compact) quantum metric spaces, and a sharp kernel obstruction for general spectral triples. This packet does not claim to resolve the source’s separate question about deriving L_p Poincaré inequalities from logarithmic Sobolev inequalities in full generality.

Source question

Marius Junge and Qiang Zeng, *Noncommutative martingale deviation and Poincaré type inequalities with applications*, arXiv:1211.3209, published in *Probability Theory and Related Fields* 161 (2015), 449–507, DOI [10.1007/s00440-014-0552-1](https://doi.org/10.1007/s00440-014-0552-1). On page 4 of the cached source version, after defining

$$\rho_L(\phi, \psi) = \sup\{|\phi(a) - \psi(a)| : L(a) \leq 1\}$$

and noting that a spectral triple has $L(a) = \|[D, a]\|$, the authors write:

“It is an interesting question but beyond the scope of this paper to determine whether the transportation inequality is possible for ρ_L .”

The Wasserstein distance has been extensively studied in the noncommutative setting; see [7, 44, 52]. This probabilistic connection which provides universal upper bound given by the entropy functional, however, seems to be new. Our definition of Wasserstein distance in the noncommutative setting is closely related to the metric used by Rieffel to define his quantum metric space. Inspired by Connes’ work in noncommutative geometry [14], Rieffel defined the metric on the state space of a $*$ -algebra A

$$\rho_L(\phi, \psi) = \sup\{|\phi(a) - \psi(a)| : L(a) \leq 1, a \in A\}$$

where ϕ, ψ are states and $L(a)$ is a seminorm. For Connes’ spectral triple, $L(a) = \|[D, a]\|$ where D is a self-adjoint operator; see [52] and the references therein for more details. It is an interesting question but beyond the scope of this paper to determine whether the transportation inequality is possible for ρ_L .

Definitions and scope

Let $(\mathcal{M}, \tau)$ be a finite von Neumann algebra with normal faithful tracial state. Let $\mathcal{A}_{\text{sa}} \subseteq \mathcal{M}_{\text{sa}}$ be a real linear unital subspace, and let $L : \mathcal{A}_{\text{sa}} \rightarrow [0, \infty)$ be a seminorm with $L(1) = 0$. For a density

$f \in L_1(\mathcal{M}, \tau)_+$ with $\tau(f) = 1$, write

$$\phi_f(a) = \tau(fa), \quad \text{Ent}_\tau(f) = \tau(f \log f),$$

with the usual value $+\infty$ allowed. Define

$$\rho_L(\phi_f, \tau) = \sup\{|\tau((f-1)a)| : a \in \mathcal{A}_{\text{sa}}, L(a) \leq 1\}.$$

The transportation inequality $T_1(C)$ means

$$\rho_L(\phi_f, \tau) \leq \sqrt{2C \text{Ent}_\tau(f)} \quad \text{for every density } f. \quad (1)$$

This is exactly the normal-state, finite-tracial setting used by the source.

Proof intuition

The carré-du-champ does no work in the source's final Bobkov–Götze duality argument: only the collection of admissible observables matters. Replacing that collection by the L -unit ball therefore gives an exact criterion for ρ_L . Entropy is convex-dual to the logarithmic moment generating function, so $T_1(C)$ is equivalent to a common subgaussian bound for all centered observables with $L(a) \leq 1$.

Finite metric diameter then supplies this bound without any semigroup or curvature hypothesis. Every admissible self-adjoint observable has spectral range at most the diameter. Tilting its spectral distribution and bounding the tilted variance by one quarter of the squared range gives Hoeffding's constant $\Delta_L^2/8$ in the logarithmic moment generating function, hence $C = \Delta_L^2/4$. In the opposite direction, a non-scalar element killed by L can be scaled at no cost and makes ρ_L infinite against an explicit bounded density.

Exact characterization and optimal constant

Define

$$C_L = \sup_{\substack{a \in \mathcal{A}_{\text{sa}}, L(a) \leq 1 \\ t \in \mathbb{R} \setminus \{0\}}} \frac{2}{t^2} \log \tau\left(e^{t(a-\tau(a)1)}\right) \in [0, \infty]. \quad (2)$$

Theorem 1 (Seminorm Bobkov–Götze duality). *For $C \in [0, \infty)$, inequality (1) holds if and only if*

$$\log \tau\left(e^{t(a-\tau(a)1)}\right) \leq \frac{Ct^2}{2} \quad (3)$$

for every $a \in \mathcal{A}_{\text{sa}}$ with $L(a) \leq 1$ and every $t \in \mathbb{R}$. Consequently C_L is the optimal T_1 constant: a finite constant exists exactly when $C_L < \infty$, and then (1) holds with $C = C_L$.

Proof. Put $x = a - \tau(a)1$. Since $L(1) = 0$, the seminorm property gives $L(x) = L(a)$. The Gibbs variational formula in a finite tracial algebra is

$$\log \tau(e^h) = \sup_{\substack{f \geq 0 \\ \tau(f)=1}} \{\tau(fh) - \text{Ent}_\tau(f)\} \quad (4)$$

for bounded self-adjoint h . Indeed, the inequality “ $\geq$ ” follows by taking $f = e^h/\tau(e^h)$, while the reverse inequality is positivity of the relative entropy of f with respect to that Gibbs density. Truncation gives the same formulas for all finite-entropy L_1 densities needed below.

Assume (3). Formula (4), with $h = tx$, gives for $t > 0$

$$t\tau(fx) \leq \text{Ent}_\tau(f) + \frac{Ct^2}{2}.$$

Optimizing in t yields $\tau(fx) \leq \sqrt{2C \text{Ent}_\tau(f)}$. Applying this to $-x$ and taking the supremum over the L -unit ball proves (1).

Conversely, suppose (1). For $L(a) \leq 1$ it implies

$$t\tau(fx) - \text{Ent}_\tau(f) \leq t\sqrt{2C \text{Ent}_\tau(f)} - \text{Ent}_\tau(f) \leq \frac{Ct^2}{2}$$

for $t \geq 0$, because the supremum of $t\sqrt{2Cs} - s$ over $s \geq 0$ is $Ct^2/2$. Taking the supremum over f in (4) proves (3) for $t \geq 0$; replacing a by $-a$ handles $t < 0$. The statement about C_L now follows directly from (2). $\square$

Finite diameter gives a universal positive answer

Let $S(\mathcal{M})$ denote the state space and set

$$\Delta_L = \text{diam}(S(\mathcal{M}), \rho_L) = \sup_{\phi, \psi \in S(\mathcal{M})} \rho_L(\phi, \psi).$$

Theorem 2 (Diameter theorem). *If $\Delta_L < \infty$, then*

$$C_L \leq \frac{\Delta_L^2}{4} \quad \text{and hence} \quad \rho_L(\phi_f, \tau) \leq \Delta_L \sqrt{\frac{\text{Ent}_\tau(f)}{2}}$$

for every density f .

Proof. Fix $a = a^*$ with $L(a) \leq 1$. If $m = \min \sigma(a)$ and $M = \max \sigma(a)$, evaluation on states attaining the spectral extrema gives

$$M - m = \sup_{\phi, \psi \in S(\mathcal{M})} |\phi(a) - \psi(a)| \leq \Delta_L.$$

Let $x = a - \tau(a)1$ and $H(t) = \log \tau(e^{tx})$. Functional calculus reduces this to the logarithmic moment generating function of the scalar spectral measure of x under τ . Thus $H(0) = H'(0) = 0$, and

$$H''(t) = \text{Var}_{\mu_t}(x) \leq \frac{(M - m)^2}{4} \leq \frac{\Delta_L^2}{4},$$

where μ_t is the exponentially tilted spectral measure. The variance bound follows from $\text{Var}(X) \leq (M - \mathbb{E}X)(\mathbb{E}X - m) \leq (M - m)^2/4$ for every random variable supported in $[m, M]$. Integrating twice, for positive or negative t , gives

$$\log \tau(e^{tx}) = H(t) \leq \frac{\Delta_L^2 t^2}{8}.$$

Theorem 1 applies with $C = \Delta_L^2/4$. $\square$

Corollary 3 (Compact quantum metric spaces). *Suppose L is a Rieffel Lip-norm on the self-adjoint part of a unital C^* -algebra represented in $(\mathcal{M}, \tau)$, so that ρ_L metrizes the weak-star topology on the state space. Then $\Delta_L < \infty$ and the transportation inequality in Theorem 2 holds. In particular, the answer is positive for every finite-tracial spectral triple for which $L(a) = \|[D, a]\|$ is a Lip-norm.*

Proof. The state space is weak-star compact. If ρ_L metrizes that topology, it is a compact metric space and therefore has finite diameter. Apply Theorem 2. $\square$

The kernel obstruction

Theorem 4 (Failure for degenerate spectral triples). *If there is a non-scalar $a = a^* \in \mathcal{A}_{\text{sa}}$ with $L(a) = 0$, then there exists a bounded, strictly positive density f with finite entropy such that*

$$\rho_L(\phi_f, \tau) = \infty.$$

Consequently no finite-constant transportation inequality can hold.

Proof. Set $b = a - \tau(a)1$. Then $b \neq 0$, $\tau(b) = 0$, and $L(b) = 0$. Choose a nonzero real ε with $|\varepsilon| < \|b\|_\infty^{-1}$ and put $f = 1 + \varepsilon b$. Then f is a bounded strictly positive density, so $\text{Ent}_\tau(f) < \infty$. For every $s \in \mathbb{R}$, the observable sb is admissible because $L(sb) = 0$, while

$$|\tau((f - 1)sb)| = |s\varepsilon| \tau(b^2).$$

This tends to infinity with $|s|$, proving the claim. $\square$

Verification and relation to the source

The proof has four independent checks.

1. The two directions of Theorem 1 use the same Gibbs variational identity, so the constants can be checked by the scalar maximization $\sup_{s \geq 0} (t\sqrt{2Cs} - s) = Ct^2/2$.
2. The diameter constant is checked by the extremal two-point distribution: the universal centered log-moment bound for range length Δ is $\Delta^2 t^2/8$, exactly corresponding to $C = \Delta^2/4$.
3. The kernel witness is explicit, positive, normalized, and has finite entropy; scaling the zero-seminorm observable forces infinite distance.
4. For a Rieffel Lip-norm, compactness of the state space immediately gives finite diameter, so no unstated semigroup, curvature, or diffusion hypothesis enters Corollary 3.

Junge–Zeng prove a Bobkov–Götze equivalence for their specific carré-du-champ unit ball (their Theorem 3.17) and a finite-diameter criterion for that same semigroup metric (their Corollary 3.19). The new step here is to observe and prove that the variational mechanism is completely independent of the carré du champ, apply it to the separately posed Connes–Rieffel metric, identify the optimal constant (2), obtain the diameter constant $\Delta_L^2/4$, and pair it with the exact kernel obstruction.

Novelty search and limitations

The run’s solution, attempt, registry, and literature indexes were searched for arXiv:1211.3209 and the phrases “Rieffel metric”, “Connes metric”, “transport entropy”, and “spectral triple”; no duplicate result was found. A bounded arXiv search used the combinations “Rieffel metric transport entropy inequality spectral triple”, “transportation inequality compact quantum metric”, and “rho_L entropy transportation inequality”. It found work establishing that various Connes seminorms are compact quantum metrics, including Rieffel’s group C^* -algebra examples and later spectral-metric examples, but no exact later paper resolving the source’s entropy inequality.

The result is mathematically elementary once the correct dual formulation is isolated, so originality confidence is lower than validity confidence; a reviewer should run a broader citation search before

public novelty claims. The theorem concerns normal states with densities relative to a finite trace, exactly the source’s probabilistic framework. It does not define entropy for arbitrary singular states, does not assert finite diameter for every spectral triple, and does not resolve the distinct LSI-to- L_p Poincaré question.

Human-review recommendation

Likely valid; promote after checking scope and novelty. The formal proof is short and constant-exact. The most important review points are (i) retaining the finite-tracial normal-state scope in any public statement, (ii) confirming that the intended meaning of “possible” in the source accepts an exact criterion plus the compact-quantum-metric corollary, and (iii) checking whether the seminorm-agnostic abstraction of the source’s Theorem 3.17 has appeared elsewhere.

References

- [1] M. Junge and Q. Zeng, *Noncommutative martingale deviation and Poincaré type inequalities with applications*, Probab. Theory Related Fields 161 (2015), 449–507, arXiv:1211.3209.
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